

Baseline Models for BatFormer

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This document explains the baseline models that are compared to our implementation of the Spacetimeformer (Grigsby et al. 2021) for bat next chirp prediction.

Note on the structure of this package:

The package of baseline models is structured as follows:

```
None
| src
| - | baseline_models
| - | - __init__.py
| - | - global_mean_predictor.py
| - | - informer.py
| - | - local_mean_predictor.py
| - | - repeat_last_predictor.py
| - | - run_baseline.py
```

There are four classes that contain baseline models (global_mean_predictor.py, informer.py, local_mean_predictor.py, repeat_last_predictor.py). run_baseline.py is the command-line script to run the baseline models, with an argument to pick which model to use. The script will train and test the model, outputting the MSE loss (although the format for the informer is a little different).

Repeat Last

The Repeat Last baseline predicts the next chirp by repeating the last chirp in the context sequence. The motivation for this is that consecutive chirps in a sequence are not likely to vary drastically from one to the next. This model has no training step, as predictions are done purely from the local context.

From the project root (bats), run

```
Shell
python src/baseline_models/run_baseline.py \
    --dataset_path datasets/2022_barn_Myca_quantile/splits \
    --model repeat-last \
    --random_seed 31
```

Global Mean

The Global Mean baseline predicts the next chirp to be the mean of the chirp attributes from the training set. This means every next chirp prediction is the same. Because the dataset may be very large, this baseline has an additional parameter that is the sample size, or the number of chirps to be used to find an approximate global mean.

From the project root (bats), run

Shell

```
python src/baseline_models/run_baseline.py \
  --dataset_path datasets/2022_barn_Myca_quantile/splits \
  --model global-mean \
  --random_seed 31
  --sample_size 5000
```

Local Mean

The Local Mean baseline predicts the next chirp to be the mean of the chirp attributes from the context sequence. The motivation for this is similar to Repeat Last in that chirps within a sequence are likely to be similar to one another, but it is not so strict in that this model uses the entire context to build a “profile” of what chirps in the sequence are like. This model also has no training step, as predictions are done purely from the local context.

From the project root (bats), run

Shell

```
python src/baseline_models/run_baseline.py \
  --dataset_path datasets/2022_barn_Myca_quantile/splits \
  --model local-mean \
  --random_seed 31
```

Informer

The Informer is the strongest baseline (see Hugging Face’s documentation [here](#) and blog post [here](#)). The Informer is a transformer-based multi-variate time series model, essentially acting as a simpler version of our Spacetimeformer. The implementation in this package is not particularly fine-tuned, which means that the model that we train for our bat chirps should definitely perform better than this model.

The outputs of this model are different from the other three baselines. It currently does not print out the loss as the others do, but instead saves files containing its predictions, ground truths,

and confidences (which mirrors the way that the Spacetimeformer outputs results). You can use those to calculate an aggregate loss value to compare between different models.

From the project root (bats), run

Shell

```
python src/baseline_models/run_baseline.py \
  --dataset_path datasets/2022_barn_Myca_quantile/splits \
  --model informer \
  --random_seed 31 \
  --n_epochs 10 \
  --seq_len 21 \
  --model_path models/informer_2022_barn_Myca_quantile
```

This will output a folder containing .log (essentially .csv) files containing the predictions, ground truths, and confidences for each model.

None

```
| models
| - | informer_2022_barn_Myca_quantile
| - | - | plots
| - | - | - loss.png
| - | - config.json
| - | - model.safetensors
| outputs
| - | informer_2022_barn_Myca_quantile
| - | - model_confidences.log
| - | - model_ground_truths.log
| - | - model_predictions.log
```

Command-line arguments

From run_baseline.py

--dataset_path DATASET_PATH

Path to the dataset

--model MODEL

Which type of baseline ["global-mean", "local-mean", "repeat-last", "informer"]

--random_seed RANDOM_SEED

Random seed for dataset

--sample_size SAMPLE_SIZE

Size of sample (instead of using whole population) for global mean

From Informer

--n_epochs N_EPOCHS
 Number of epochs to train
--seq_len SEQ_LEN
 Length (padded) of the input sequences
--model_path MODEL_PATH
 Filepath to save models